

RAIL SW DEV AND V&V ENGINEER

7+ YEARS OF EXPERIENCES



Knowledge and Expertise

- ✓ 7+ years of experience in Software Verification and Validation, System Engineering and Software Development, under Rail Transportation and Medical domains,
- ✓ Test plan development, test procedures and reports redactions,
- ✓ Lab testing expertise,
- ✓ Requirements gathering, user story development, software design, programming languages (e.g., Java, Python, C#)
- ✓ Automated test plans, unit testing, dynamic application security testing, interface testing, integration testing, end-to-end testing, user acceptance testing
- ✓ Cross-functional team collaboration, stakeholder communication, engineering and architecture assessment
- ✓ Software deployment, configuration and environment troubleshooting, code fix implementation
- ✓ Testing and commissioning on the trackside CBTC signaling system, Rolling Stock(TCMS)
- ✓ Experiences with C, C#, C++, Python and Linux environment,
- ✓ Good knowledge TCMS and ATS, Rail Standards
- ✓ Authoring/executing Testing & Validation documentation includes Test/Validation Plan, audit plans, requirement specifications, defining Validation Strategy, Validation Protocol (IQ, OQ and PQ), Summary Report, Test Estimations, Test Strategy, Test Cases/Scenarios, Traceability Metrics and Validation reports and Test documentation

- ✓ **Collaboration with Business Owners with different departments to ensure CSV and compliance with FDA requirements while monitoring quality metrics, performing risk assessments, and defining SOPs**
- ✓ **Good knowledge on using MITRAC tools (MTPE, MWT, MDCT, BT-SCT, Bombardier Proprietary - C# Testbench, Virtual Train), RCS Tools**
- ✓ **Experience in Software Development activities**
- ✓ **Experience in statistical data analytics, data modeling, data visualization**
- ✓ **Thorough Understanding of Software Development Life Cycle (SDLC), involved in various phases like**
- ✓ **Requirements, Analysis/Design, Development and Testing as Technical Assurance Analyst to ensure adherence to compliance and SOPs.**
- ✓ **Actively participated in Requirement Gathering Calls with the Business and the Technical team.**
- ✓ **Lead the Defect Call for all the previous testing Project with the Business and the Technical team**

Technical Skills

- **Programming Languages: C, C#,C++ Python, SQL,Shell Scripting,Perl**
- **Data Visualization tools: Power BI, IBM SPSS Modeler, Tableau**
- **Development Tools: Visual Studio 2010/2012/2013, QT-Creator**
- **Methodologies: SDLC, Agile, Waterfall, V-Module**
- **Languages: Java, JavaScript (React.js), SQL**
- **Tools: Selenium (for automated testing), JIRA (for project management and issue tracking), Jenkins (for continuous integration and deployment), Docker (for containerization)**
- **Testing Frameworks: JUnit (for unit testing), OWASP ZAP (for dynamic application security testing)**
- **Protocols: CAN, UART, I2C, SPI,BLUETOOTH,IP ETHERNET**
- **Bombardier Test Tools: MTPE, MWT, Safeprog, BT Test framework, Virtual Train, MTF, DCU Term, RTSIM**
- **Configuration Management Tools: RTC, SVN, SPARKX - ENTERPRISE ARCHITECT**
- **Bug Reporting Tools: JIRA, RTC(JAZZ),IBM(DOORS),ENTERPRISE ARCHITECT(SPARKX)**
- **IDE's: IAR, MPLABX, Microsoft Visual Studio**

WORK EXPERIENCE

Oct 2023 – current (1+ YEAR)	
<u>Senior Software Engineer(ITC Test & Commissioning)</u>	IKOS Consulting – Consultant for Alstom – DALLAS FORTWORTH INTERNATIONAL AIRPORT <i>Dallas, TX (onsite)</i>
Project: CBTC Upgrade for the DFW APM System	
Activity & Responsibility: ○ Collaborated with stakeholders including product managers, business analysts, and end-users to gather and review detailed software requirements and user stories. ○ Created estimates for development tasks and translated them into actionable software design specifications. ○ Worked closely with engineers and architects to assess and test hardware and software interactions, ensuring compatibility and performance standards were met ○ Executed an Agile development methodology, participating in sprints and daily stand-ups to ensure timely delivery and alignment with project goals. ○ Applied various programming languages (e.g., Java for backend services, JavaScript for frontend development) to develop and implement new features and functionalities ○ Developed and executed automated test plans and strategies utilizing business requirements. ○ Collaborated with engineers and architects to validate test environments, ensuring they accurately reflected production settings. ○ Created and maintained test data, executed test cases, and validated results against expected outcomes. ○ Designed and implemented code fixes for issues identified during testing, produced detailed test reports to communicate findings and progress to stakeholders ○ Field Testing and Commissioning, day shifts, night shifts, weekend shifts, ○ Installation & integration of the system: PICO (Post Installation CheckOut), Replace control systems and control equipment, verify the equipment has been correctly installed and connected, ○ Static testing (SAT) on CBTC system & validate wayside equipment without train running, ○ Conducted various types of testing including unit testing, dynamic application security testing to identify vulnerabilities, interface testing to ensure smooth interaction between components, integration testing for combined functionality, end-to-end testing to validate the entire application workflow, and user acceptance testing (UAT) to ensure the solution met user needs ○ Dynamic testing on OCS, station doors, braking, alarms, ... Supported applications by resolving configuration and environment issues, ensuring reliability. ○ Facilitated the software deployment process, coordinating with operations teams to manage releases and ensure smooth transitions to production environments ○ SW testing & safety testing.	

Sep 2022 – Sep 2023	
<u>Validation and Test Engineer</u>	CVS Health <i>Pennsylvania(Internship)</i>
Project 1: QA Automation Activity & Responsibility: ○ Wrote Linux based scripts to automate test cases using batch to test speech recognition on tablets using HMI automation ○ Executed Black box testing and monitor debug nightly automated test runs to identify the root cause audio, video and HDMI failures on smart phones and tablet per daily basic usage of emulators and ADB commands and Linux environment, ○ Performed configurations of hardware such as intel, NVIDIA and ATI GPU cards and software on test machines (Win2K, WinXP, Vista, Mac-OS, and linux red hat 4.0/SUSE 10.1 systems), ○ Submit Test cases to Client team members for Peer-to-Peer Review, ○ Review and Coordination of Test Case execution, ○ Design Change Implementation Plan, Design Verification document, Change Implementation Report and RTM, ○ Execute the Design Verification, Regression and Sanity test cases, ○ Retesting of defects raised during Design Verification and Regression, ○ Worked on paper-based execution and in the later phase used HP ALM for Design Verification phase, ○ Worked on creating FAQ documents, SOP for Training Support team, eLearning before the launch of mobile APP. Designed test cases based on the use-cases and the complex business logic defined for each web-service or a message.	

Aug 2019 – Aug 2022 (3 years)	
<u>Verification and Validation Test Lead</u>	Consultant for Bombardier/Alstom <i>India</i>
Project 1: Lead SW/HW Integration Test Engineer - Automatic Train Supervision Control Activity & Responsibility: ○ Mentoring as lead integrator for transmission and reception of CAN messages from ATS device to wayside and wayside to ATS device, ○ Test automation using python – developed, maintained test cases for smooth testing in 2F modules, ○ Defined test scenarios, created test plans, and executed test cases to automate the integration functionality of software-to-software modules and software to hardware modules, ○ Install, configure, and run modelling and simulation software, ○ Generate and maintain test procedures and processes in C# for installation and configuration of software to be integrated using C# Test bench,	

- Perform failure analysis using Alstom specific tools, validate scripts and provide these findings to stakeholders,
- Work within agile team environments with a multi - disciplinary team of engineers.
- Peer reviewed Requirements plan, Test plan and generated FVRDS,SRS,ICD
- Reviewed derived requirements, including safe, non-safe, Functional and Non functional requirements.
- Performed Requirement Analysis at component level, integration and vehicle level
- Performed Risk Analysis – Identified gaps, redundancies, Generated KPI reports, RCA's,Review checklists, Artifacts
- Prepared weekly, monthly status reports to customer
- Performed management activities with Quality team, cross functional teams along with active technical activities.

Project 2: System Test Engineer – Train Control Management System – LRV Brussels

Activity & Responsibility:

- Mentoring teammates on Functional Requirement Analysis, Software Design, Software Requirement Analysis, Code implementation,
- Underwent the designing to test automated framework,
- Designed the test automated framework in implementing reusable components and integrating the test automation scripts into the overall functional requirements of system modules,
- Developed Software Test Plans
- Developing the test cases in C# in Enterprise Architect tool,
- Developing the test cases based on Functional Requirements,
- Generation of Test Specification Reports C# using Enterprise Architect tool,
- Developing the test scripts using C# platform,
- Execution of test cases on Virtual Train platform,
- Reviewing the Test Scripts,
- Updating checklists, dashboards,
- Raising bugs and analyzing the test issues,
- Client Discussions, sending weekly reports to customer, Updating the KPI's,
- Developing the Software Design Documents basing on Functional Requirement Specification,
- Involved in developing software design based on System Requirements,
- Code implementation for the Software Requirement Specification,
- Analyzing the Functional Requirement Specification,
- Conducted trainings in sharing knowledge on TCMS System Architecture, System Design and Module Design.

Project 3: System Test Engineer – Train Control Management System – LRV Zurich

Activity & Responsibility:

- Provided technical support for test automation team of 20 members – like standards of test automation solutions using PYTHON, test automation solutions for infotainment modules,
- Mentoring teammates on Functional Requirement Analysis, Software Design, Software Requirement Analysis, Code implementation,
- Underwent internal discussions on design analysis (module design and system design),
- Developed Software Test Plans,
- Involved in internal audits and responsible in generation of final project reports,
- Generation of Test Specification Reports using Enterprise Architect tool,
- Developing the test scripts using C# platform,
- Execution of test cases on Virtual Train platform,
- Reviewing the Test Scripts,
- Updating checklists, dashboards,
- Raising bugs and analyzing the test issues,
- Estimating the given inputs, impact analysis for the given inputs,
- Client Discussions, sending weekly reports to customer, Updating the KPI's,
- Peer reviewing the work tasks (maintaining review checklists and submitting the deliverables on time to the customer.

Sep 2017 – Aug 2019 (1 year and 11 Months)

Software Engineer

Thermofisher Scientific
India

Project: Delta K Direct. The objective of this project is to analyze the percentage of ozone in the atmospheric gases.

Activity & Responsibility:

- Developing the Graphical User Interface screens using PyQt5 for the required sprints,
- Developing new Test cases, and as well as maintaining and supporting existing Testcases,
- Performing unit level as well as bench level testing for ULT's,
- Involved in modifying business logic need functional requirements,
- Involved in updating the system requirements,
- Involved in validating graphical user interface screens on ULT.

EDUCATION

2023	Master’s Degree in Business Analytics University of New Haven
2017	Bachelor of Technology – Electronics and Communications Engineering Jawaharlal Nehru Technological University

LANGUAGES

English: Professional